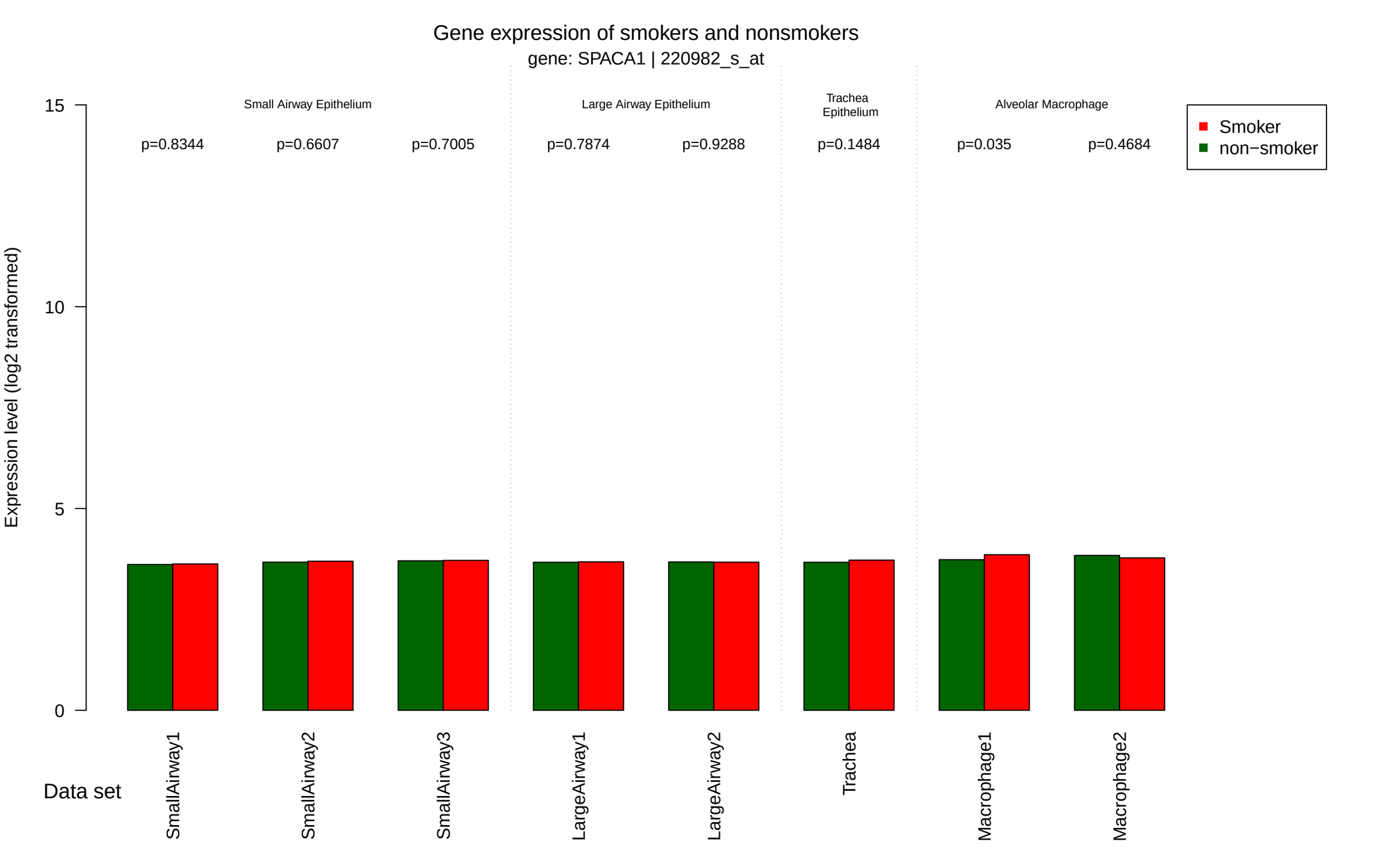
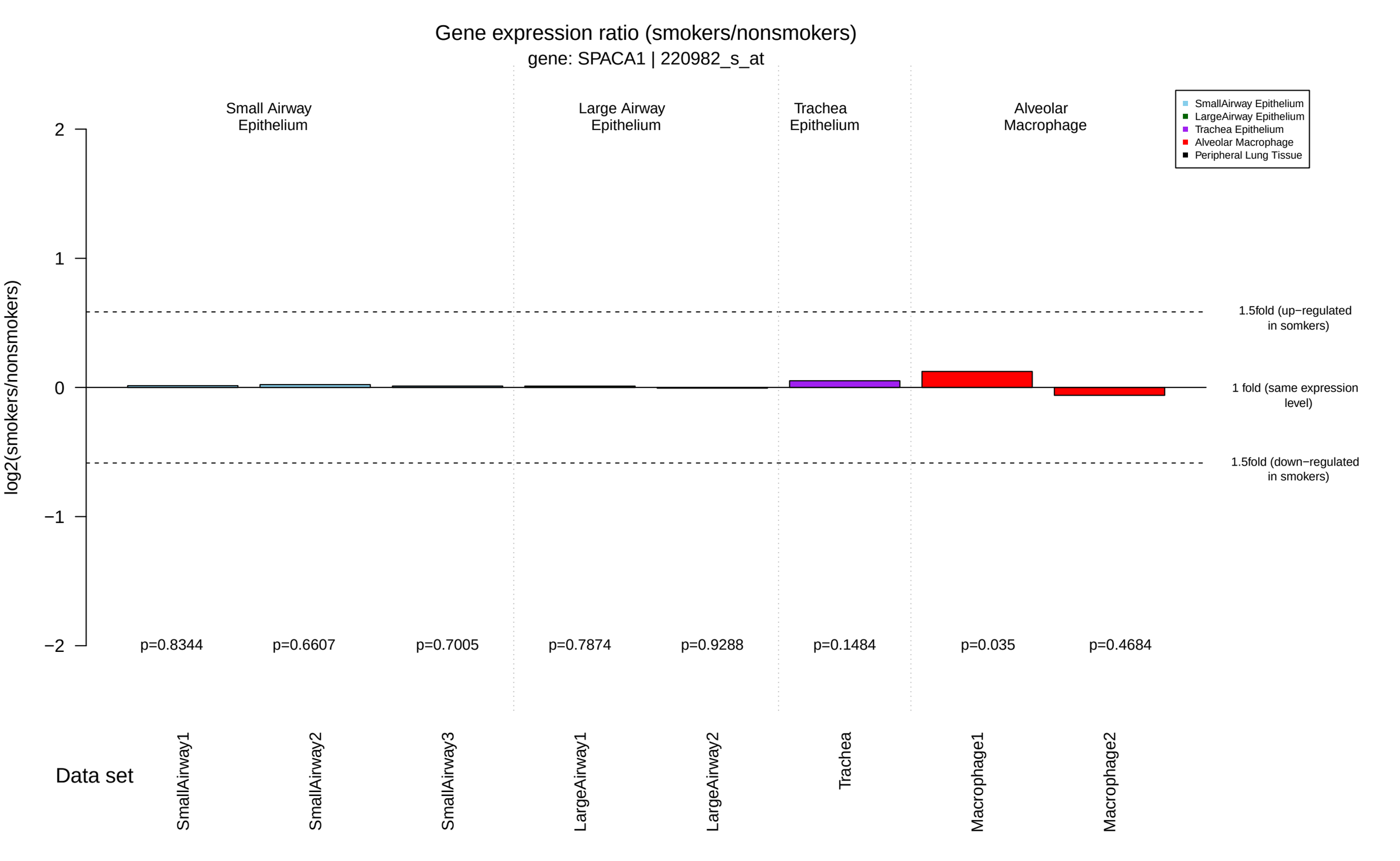
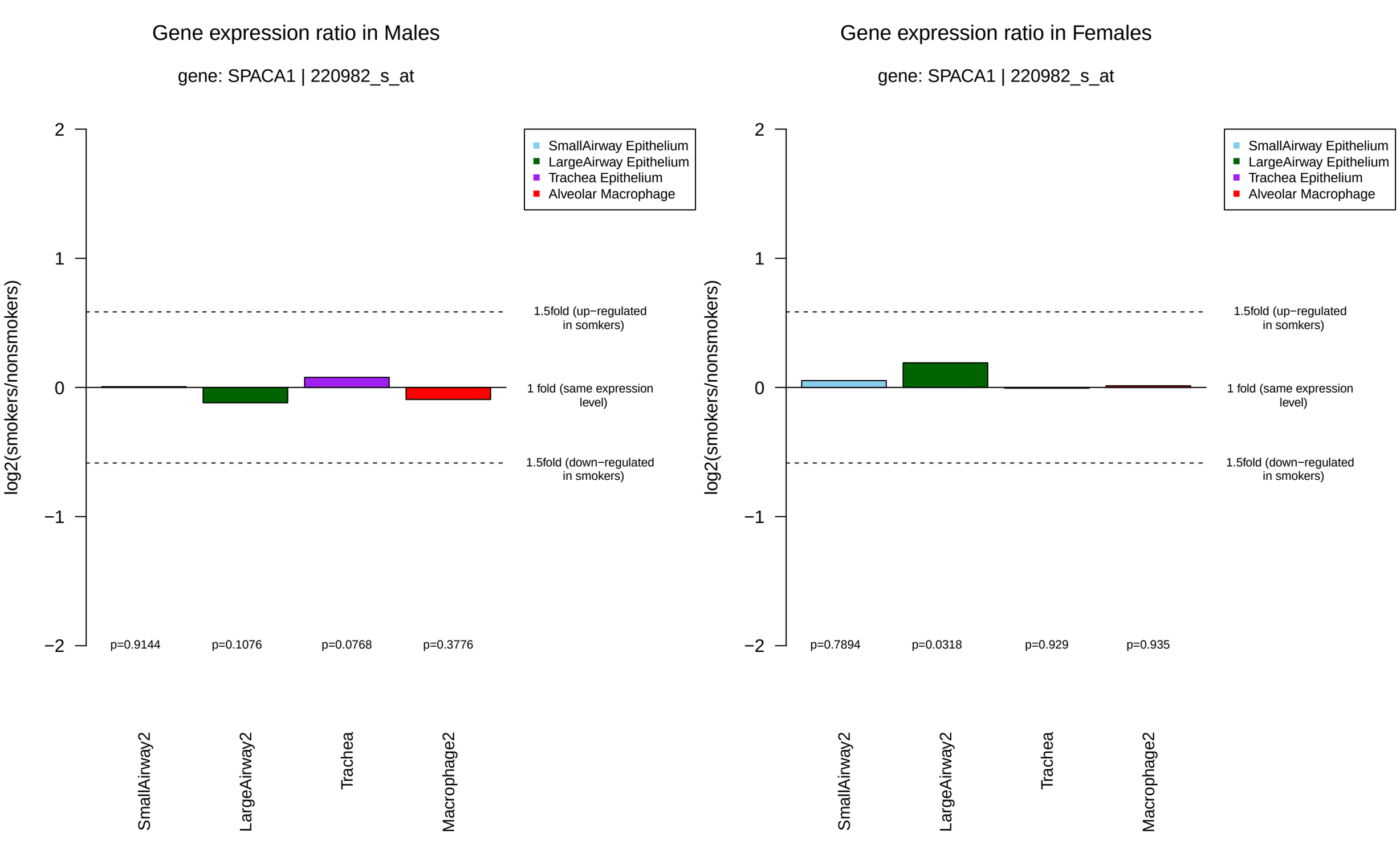
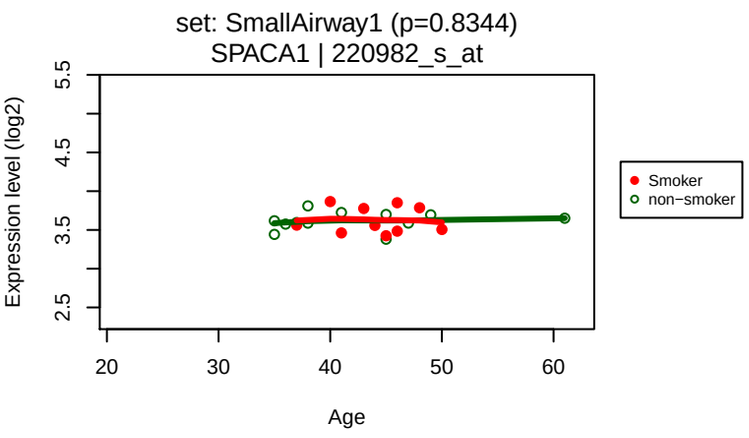
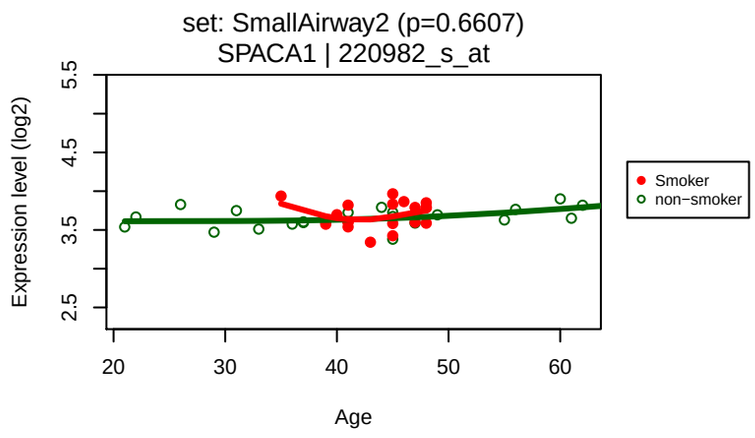
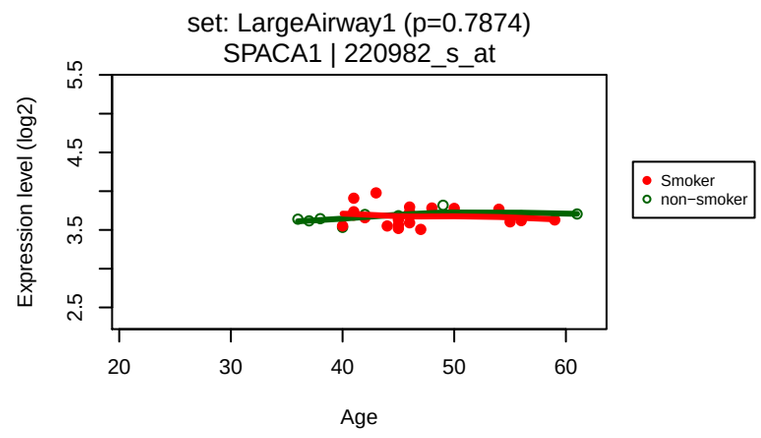

Gene expression of smokers and nonsmokers
gene: SPACA1 | 220982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8344
p=0.6607
p=0.7005
p=0.7874
p=0.9288
p=0.1484
p=0.035
p=0.4684
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

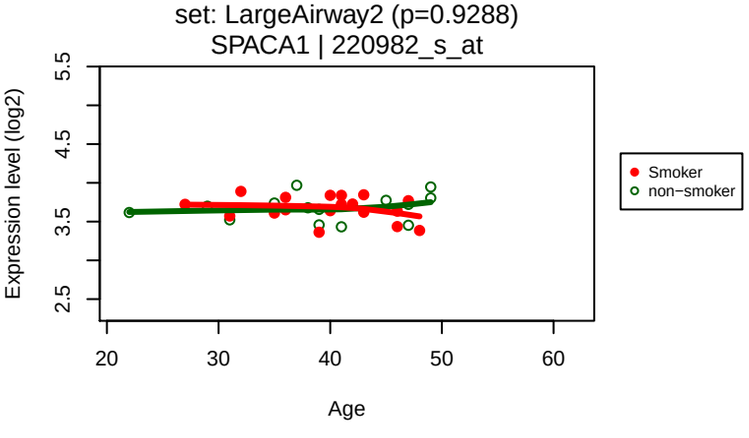

Gene expression ratio (smokers/nonsmokers)
gene: SPACA1 | 220982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8344
p=0.6607
p=0.7005
p=0.7874
p=0.9288
p=0.1484
p=0.035
p=0.4684
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

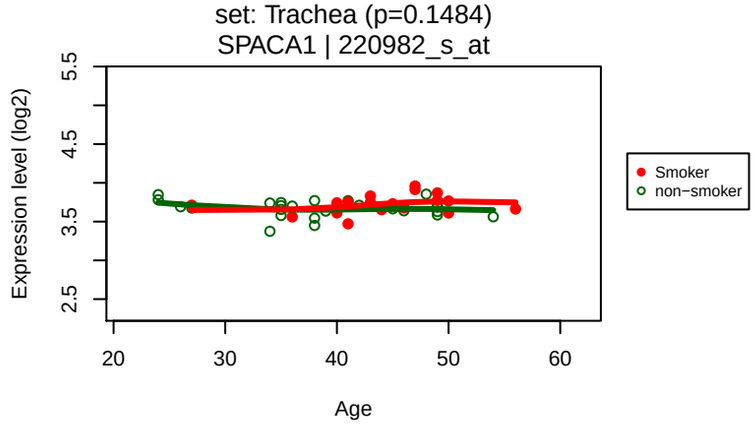

Gene expression ratio in Males
gene: SPACA1 | 220982_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9144
p=0.1076
p=0.0768
p=0.3776
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPACA1 | 220982_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7894
p=0.0318
p=0.929
p=0.935
SmallAirway2
LargeAirway2
Trachea
Macrophage2

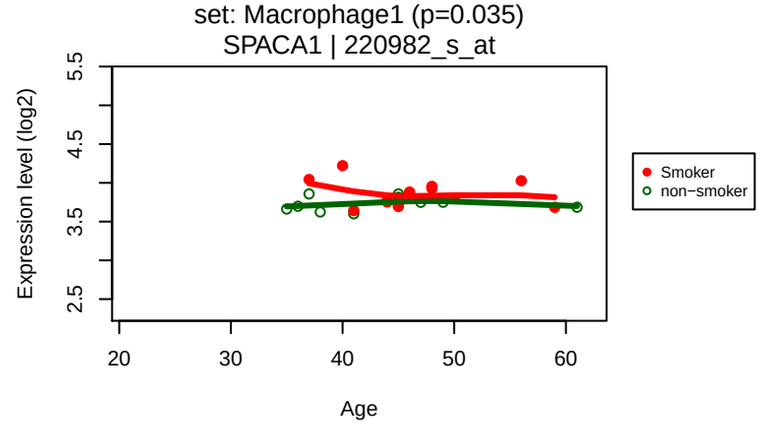

set: SmallAirway1 (p=0.8344)
SPACA1 | 220982_s_at
Expression level (log2)
Age
Smoker
non−smoker

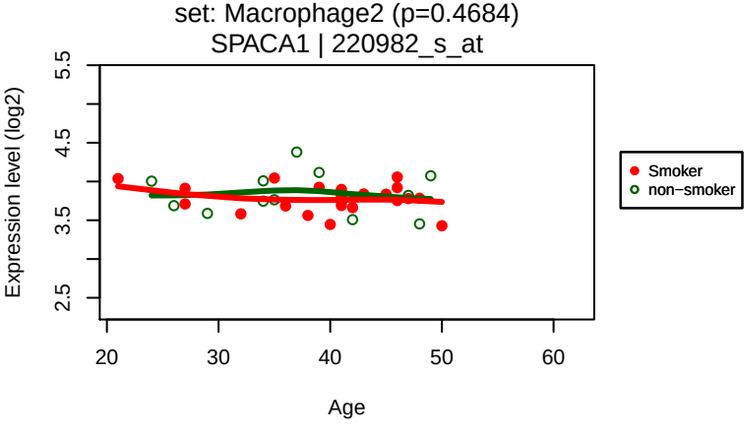

set: SmallAirway2 (p=0.6607)
SPACA1 | 220982_s_at
Expression level (log2)
Age
Smoker
non−smoker

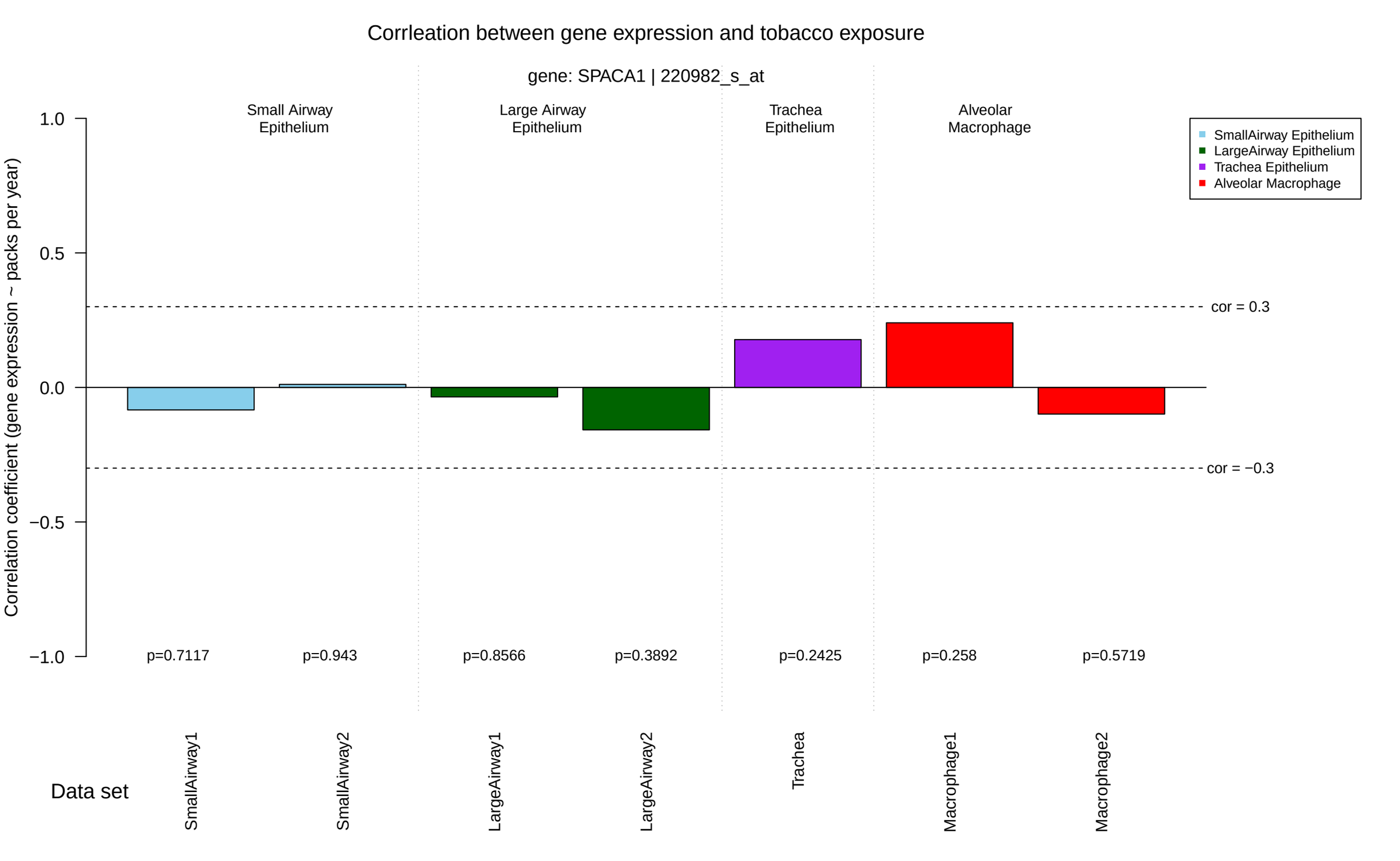

set: LargeAirway1 (p=0.7874)
SPACA1 | 220982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9288)
SPACA1 | 220982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1484)
SPACA1 | 220982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.035)
SPACA1 | 220982_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4684)
SPACA1 | 220982_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPACA1 | 220982_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7117
p=0.943
p=0.8566
p=0.3892
p=0.2425
p=0.258
p=0.5719
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2